

# Lab Sheet 08

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**Title:** Smart Garbage Bin Weight Monitoring System using ESP32, HX711 & Blynk.

**Aims:**

To design and implement a smart garbage bin monitoring system using an ESP32 microcontroller and HX711 load cell amplifier, where the bin weight is continuously measured and displayed on an I2C LCD and the Blynk IoT platform. The system provides visual alerts using an RGB LED to indicate garbage levels (low, medium, full) and detects overweight conditions, thereby optimizing waste collection efficiency.

**Task 01:** Smart Home Automation System Setup

1. Project Setup in Wokwi / VS Code
  - a. Name the project **SmartGarbage\_ESP32\_Blynk**.
2. Circuit Design
  - a. Sensor Connection
    - i. Connect the HX711 Load Cell Amplifier to the ESP32.

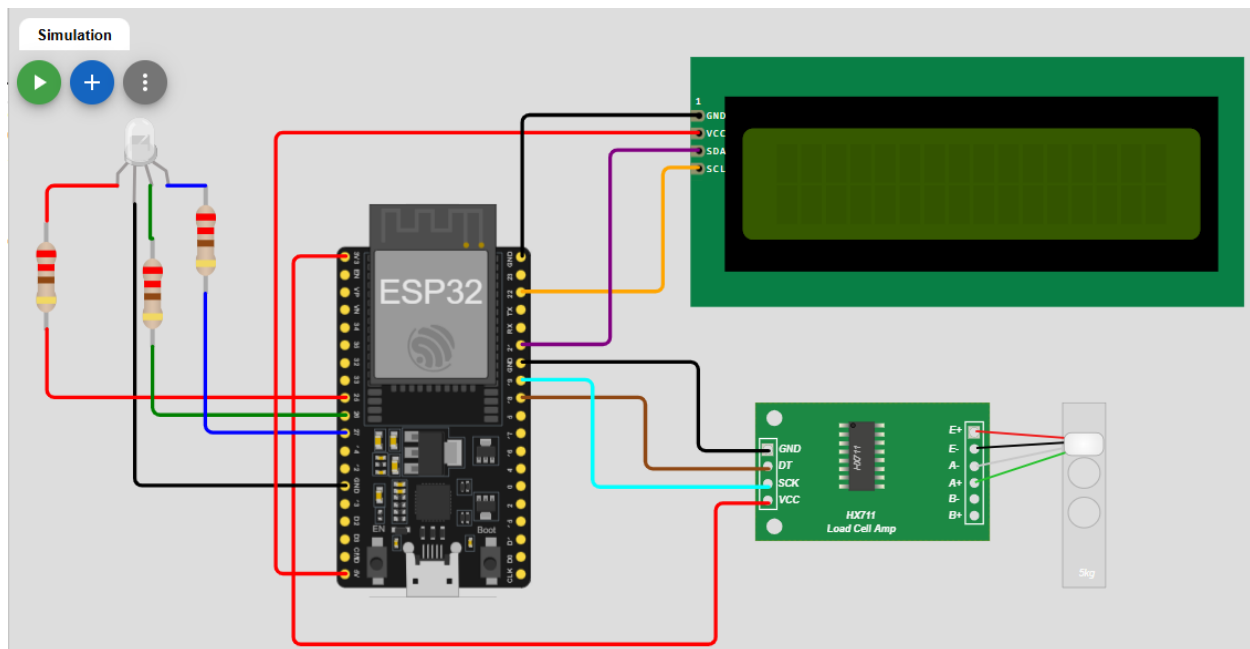
| HX711 | ESP32   |
|-------|---------|
| DT    | GPIO 18 |
| SCK   | GPIO 19 |
| VCC   | 3.3V    |
| GND   | GND     |

- a. Output Devices Connection
    - i. Connect RGB LED and LCD(I2C) display to the ESP32.

| LCD | ESP32   |
|-----|---------|
| SDA | GPIO 21 |
| SCL | GPIO 22 |

|     |     |
|-----|-----|
| VCC | 5V  |
| GND | GND |

| RGB LED        | ESP32   |
|----------------|---------|
| Red            | GPIO 25 |
| Green          | GPIO 26 |
| Blue           | GPIO 27 |
| Common Cathode | GND     |

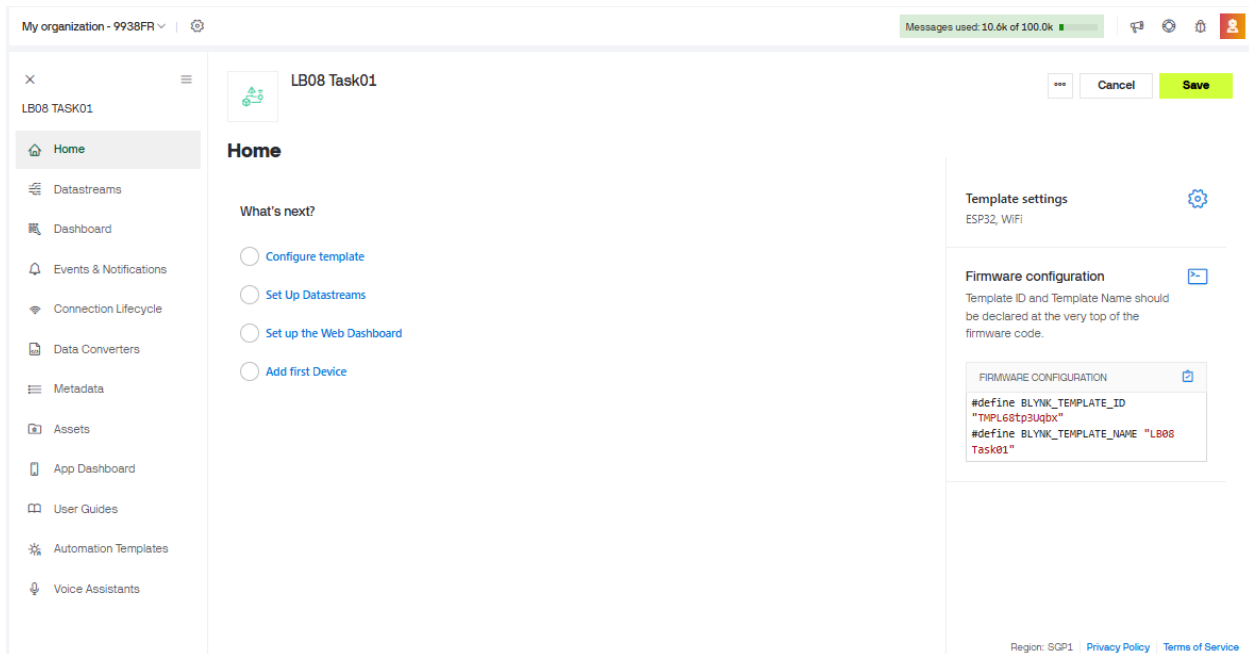


### 3. Library Integration

```
[env:esp32dev]
platform = espressif32
board = esp32dev
framework = arduino
lib_deps =
    blynkkk/Blynk@^1.3.5
    bogde/HX711@^0.7.5
    marcoschwartz/LiquidCrystal_I2C@^1.1.4
```

## 4. Blynk Configuration

### a. Create a New Template in Blynk (IoT platform)



### b. Assign virtual pins

#### Create Datastream

- Name: **Weight**
- Virtual Pin: **V0**
- Data Type: **Double**
- Min: **0**
- Max: **10** (or according to your bin capacity)
- Unit: **kg**

#### Widget

Add a:

- Gauge Widget or
- Value Display Widget

Connect it to **V0**.

## New Datastream

### General

Use in Automations

Advanced Settings

Code

NAME

Weight

PIN

V0

DISPLAY NAME

Double V0

SEMANTIC TAGS (OPTIONAL)

+ Add

Help us recognize the meaning of your data (e.g., temperature, battery, RSSI) to power smarter dashboards and analytics.

DATA TYPE

Double

e.g., 5.5

UNITS

Kilogram, kg

Cancel

Save

## V1 – Bin Status (String)

### Create Datastream

- Name: **Bin Status**
- Virtual Pin: **V1**
- Data Type: **String**

### Widget

Add a:

- Label Widget
- Labeled Value Widget

Connect it to **V1**.

# New Datastream

General   Use in Automations   Advanced Settings   Code

NAME

Bin Status

PIN

V1

DISPLAY NAME

String V1

SEMANTIC TAGS (OPTIONAL)

+ Add

Help us recognize the meaning of your data (e.g., temperature, battery, RSSI) to power smarter dashboards and analytics.

DATA TYPE

String

e.g., "error" ▾

DEFAULT VALUE

Enter...

Cancel   Save

LB001 TASK01

...

Cancel

Save

Datastreams

Search datastream...

New Datastream ...

All 2

Integer

Double

String

Enum

Location

Digital

Analog

| <input type="checkbox"/> | Pin | Name       | Data Type | Semantics | Min | Max | Default | Units | Use in Automat... | Actions |
|--------------------------|-----|------------|-----------|-----------|-----|-----|---------|-------|-------------------|---------|
|                          | V0  | Weight     | Double    | + Add     | 0   | 10  | 0       | kg    | None              |         |
|                          | V1  | Bin Status | String    | + Add     | 0   | 1   |         | -     | None              |         |

- c. Create a New Device
- d. Add widgets to display values.

### Gauge Settings

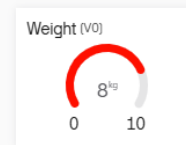
TITLE (OPTIONAL)

DATASTREAM  
 

☒ Override Datastream's Min/Max fields

LEVEL COLOR  
☒ Change color based on value





Cancel Save

### Label Settings

TITLE (OPTIONAL)

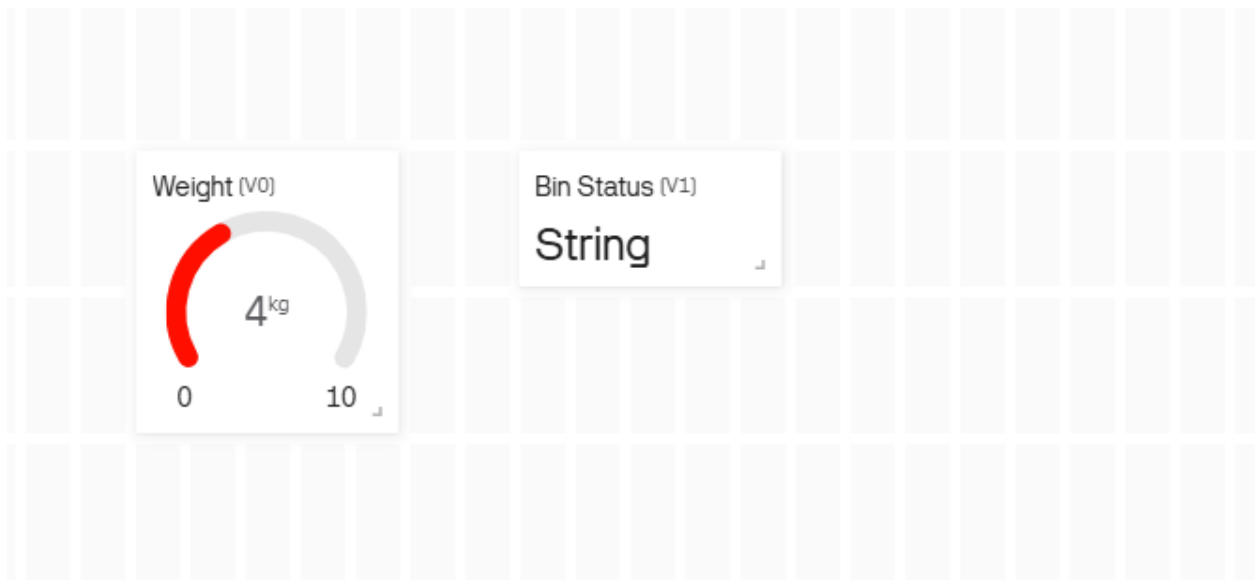
DATASTREAM  
 

CONTENT ALIGNMENT  
☒ Left ☐ Center ☐ Right

WIDGET BACKGROUND  
☐

Bin Status [V1]  
String

Cancel Save



- e. Obtain Template ID, Template Name and Auth Token

```
#define BLYNK_TEMPLATE_ID "TMPL68tp3Uqbx"  
#define BLYNK_TEMPLATE_NAME "LB08 Task01"  
#define BLYNK_AUTH_TOKEN "b-eGWP8gtiJazDjQcrCp76L1V6DfQGWF"
```

## 5. Programming Logic Development in Wokwi / VS Code

- a. Initialize Wi-Fi and Blynk connection
- b. Initialize HX711 and LCD
- c. Read weight from load cell using HX711
- d. Show weight on LCD
- e. RGB LED Logic (Level Indication)
  - i. Define thresholds:
    1. Low Level  $\rightarrow$  Weight  $< 2$  kg
    2. Medium Level  $\rightarrow 2$  kg – 5 kg

3. Full Level → > 5 kg
4. Overweight → > 7 kg ii. LED behavior:
  1. Low → Green ON
  2. Medium → Blue ON
  3. Full → Red ON
4. Overweight → Red BLINKING

```
#define BLYNK_TEMPLATE_ID "TMPL68tp3Uqbx"
#define BLYNK_TEMPLATE_NAME "LB08 Task01"
#define BLYNK_AUTH_TOKEN "b-eGWP8gtiJazDjQcrCp76L1V6DfQGWF"

#include <Arduino.h>
#include <WiFi.h>
#include <BlynkSimpleEsp32.h>
#include <HX711.h>
#include <LiquidCrystal_I2C.h>

char ssid[] = "Wokwi-GUEST";
char pass[] = "";

#define DT_PIN 18
#define SCK_PIN 19

#define RED_LED 25
#define GREEN_LED 26
#define BLUE_LED 27

HX711 scale;
LiquidCrystal_I2C lcd(0x27, 16, 2);

BlynkTimer timer;

void sendData()
{
    // Example weight for simulation
    float weight = scale.get_units();

    String status = "";

    lcd.clear();
```



```
lcd.setCursor(0,0);
lcd.print("Weight:");
lcd.print(weight);
lcd.print("kg");

// LOW LEVEL
if(weight < 2)
{
    digitalWrite(GREEN_LED, HIGH);
    digitalWrite(BLUE_LED, LOW);
    digitalWrite(RED_LED, LOW);

    status = "LOW";
}

// MEDIUM LEVEL
else if(weight >= 2 && weight <= 5)
{
    digitalWrite(GREEN_LED, LOW);
    digitalWrite(BLUE_LED, HIGH);
    digitalWrite(RED_LED, LOW);

    status = "MEDIUM";
}

// FULL LEVEL
else if(weight > 5 && weight <= 7)
{
    digitalWrite(GREEN_LED, LOW);
    digitalWrite(BLUE_LED, LOW);
    digitalWrite(RED_LED, HIGH);

    status = "FULL";
}

// OVERWEIGHT
else
{
    digitalWrite(GREEN_LED, LOW);
    digitalWrite(BLUE_LED, LOW);

    digitalWrite(RED_LED, HIGH);
    delay(300);
    digitalWrite(RED_LED, LOW);
    delay(300);
}
```

```

    status = "OVERWEIGHT";
}

lcd.setCursor(0,1);
lcd.print(status);

Blynk.virtualWrite(V0, weight);
Blynk.virtualWrite(V1, status);
}

void setup()
{
    Serial.begin(115200);

    pinMode(RED_LED, OUTPUT);
    pinMode(GREEN_LED, OUTPUT);
    pinMode(BLUE_LED, OUTPUT);

    lcd.init();
    lcd.backlight();

    scale.begin(DT_PIN, SCK_PIN);

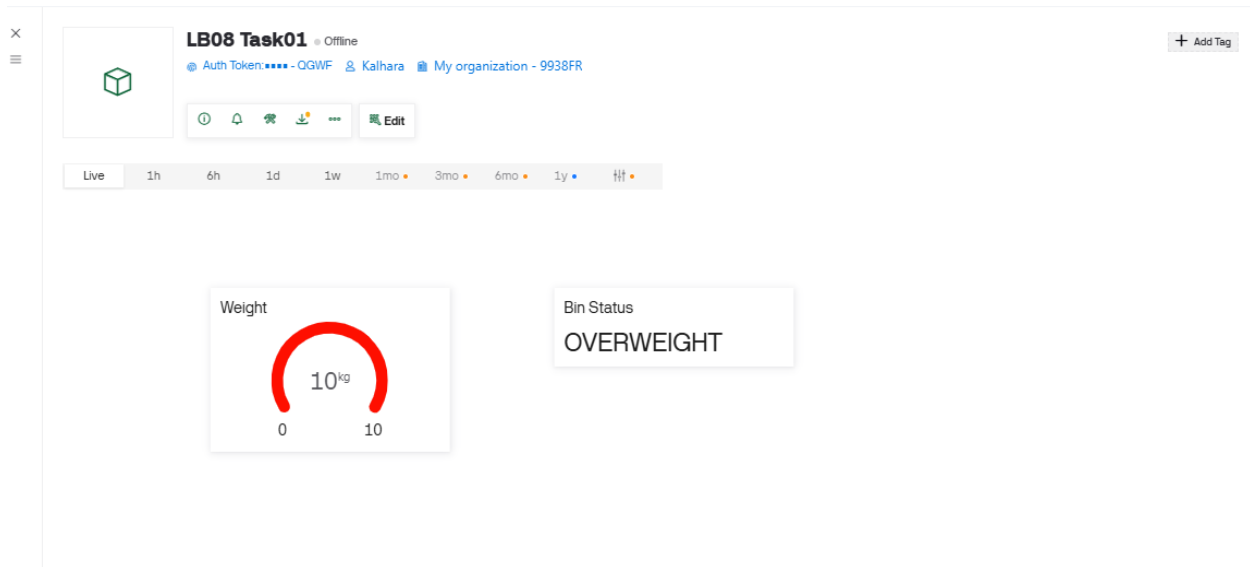
    Blynk.begin(BLYNK_AUTH_TOKEN, ssid, pass);

    timer.setInterval(2000L, sendData);
}

void loop()
{
    Blynk.run();
    timer.run();
}

```

f. Send weight data to Blynk



## 6. Simulation & Testing in Wokwi / VS Code

